

B-cell characteristics define HCV reinfection outcome

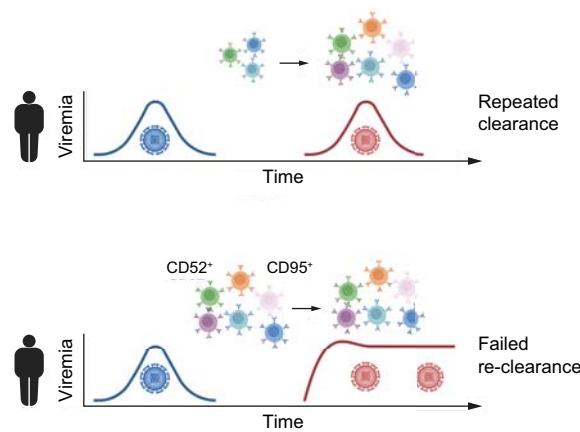
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Graphical abstract



Highlights

- Protection from persistent reinfection was associated with expansion of E2-specific B cells upon reinfection.
- E2-specific B cells have a transcriptomic profile associated with activation and proliferation prior to clearance of reinfection.
- We observed evidence of antigen imprinting in HCV reinfection.

Impact and implications

HCV continues to have a major health burden globally. Limitations in the health infrastructure for diagnosis and treatment, as well as high rates of reinfection, indicate that a vaccine that can protect against chronic HCV infection will greatly complement current efforts to eliminate HCV-related disease. With alternative approaches to testing vaccines, such as controlled human inoculation trials under consideration, we desperately need to identify the correlates of immune protection. In this study, in a small but rare cohort of high-risk injecting drug users who were reinfected multiple times, breadth of neutralisation was not associated with ultimate clearance of the reinfection event. Alternatively, characteristics of the HCV-specific B-cell response associated with B-cell proliferation were. This study indicates that humoral responses are important for protection and suggests that for genetically very diverse viruses, such as HCV, it may be beneficial to look beyond just antibodies as correlates of protection.

B-cell characteristics define HCV reinfection outcome

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Background & Aims: In individuals highly exposed to HCV, reinfection is common, suggesting that natural development of sterilising immunity is difficult. In those that are reinfected, some will develop a persistent infection, while a small proportion repeatedly clear the virus, suggesting natural protection is possible. The aim of this study was to characterise immune responses associated with rapid natural clearance of HCV reinfection.

Methods: Broad neutralising antibodies (nAbs) and Envelope 2 (E2)-specific memory B cell (MBC) responses were examined longitudinally in 15 individuals with varied reinfection outcomes.

Results: Broad nAb responses were associated with MBC recall, but not with clearance of reinfection. Strong evidence of antigen imprinting was found, and the B-cell receptor repertoire showed a high level of clonality with ongoing somatic hypermutation of many clones over subsequent reinfection events. Single-cell transcriptomic analyses showed that cleared reinfections featured an activated transcriptomic profile in HCV-specific B cells that rapidly expanded upon reinfection.

Conclusions: MBC quality, but not necessarily breadth of nAb responses, is important for protection against antigenically diverse variants, which is encouraging for HCV vaccine development.

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Introduction

In recent years, the identification of broad neutralising antibodies (BnAbs) capable of neutralising diverse viral strains has been a major focus of vaccine research for many RNA viruses.^{1–3} Passive immunisation with BnAbs has been shown to successfully protect against HCV challenge in chimpanzees and chimeric mouse models,^{4–8} and induction of BnAbs correlated with spontaneous clearance of primary infection,⁹ indicating that protection by antibodies is possible. However, in high-risk individuals, HCV reinfection rates are high,^{10–12} indicating that natural development of sterilising immunity is likely to be rare. That said, acute HCV infection is usually asymptomatic,¹³ so the burden of disease is associated with the 75% of infections that result in persistent infection. Therefore, any vaccine that can improve the natural clearance of HCV would have a significant impact on the health burden of HCV. In this regard, it is noteworthy that while only 25% of primary HCV infection episodes are naturally cleared,^{14,15} up to 80% of HCV reinfection episodes are thought to be cleared,¹⁶ implying that partial protective immunity can be induced following primary infection. Although spontaneous clearance of HCV can happen after years of infection,^{17,18} this is still associated with a higher risk of cirrhosis and hepatocellular

carcinoma.^{19,20} Therefore, investigation of factors associated with protective immunity should focus on those that rapidly clear their HCV infection.

The major target for BnAbs is the HCV envelope, which is a heterodimer formed from envelope glycoprotein 1 and 2 (E1/E2). The majority of the well-characterised antibody epitopes are found within E2,^{5,21–23} with some epitopes requiring both E1 and E2 together in a conformational state.²⁴ On E2 alone, the neutralising epitopes include antigenic region (AR)2, AR3/domain B, domain C, domain D and domain E; and non-neutralising epitopes AR1 and domain A. A recent study in an injecting drug user (IDU) cohort suggested that antibodies targeting certain epitopes, notably AR3, may confer better protection.²⁵

In recent years, the role of different memory B cell (MBC) subsets in protection against reinfections with RNA viruses, such as influenza, has garnered much interest.²⁶ The general theory is that high affinity, class-switched MBCs likely underpin the best protection against the same or highly similar antigens, as the antibodies secreted by these B cells, upon activation and further differentiation, can rapidly respond to the new viral challenge.²⁷ In the present study, HCV-specific antibody and B-cell responses were assessed longitudinally in a rare cohort of high-risk IDUs followed from unexposed status, through

Keywords: HCV; reinfection; clearance; persistence; B cell; neutralising antibody.

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natural clearance of a primary infection, and then subsequent reinfection events. To identify factors associated with clearance of reinfection, autologous and heterologous neutralisation, mapping of B-cell epitopes and characterisation of B-cell phenotypes and receptors from peripheral blood mononuclear cells (PBMCs) were assessed. Given that neutralisation of the autologous viral strain, and neutralising breadth have both previously been associated with natural clearance of HCV,^{9,28,29} it was hypothesised that these factors would also be associated with natural clearance of HCV reinfection.

Materials and methods

Please see the supplementary information and CTAT table for detailed materials and methods.

Results

Neutralising breadth in plasma prior to, and during, reinfection does not associate with outcome

Stored clinical data, as well as collected plasma and PBMCs for 15 individuals from a prospective cohort of high-risk IDUs were selected for analyses on the basis of initial HCV-negative antibody and PCR status before primary infection resulting in natural clearance, and then at least one reinfection episode (Table 1). For these 15 individuals, a total of 24 reinfection events (39 total infection events including primary infection) were documented with each event being defined as resulting in clearance (C, n = 16), or persistence (P, n = 8) depending on the duration of viraemia (>365 days defined as persistence). For some individuals, multiple reinfection events resulted in clearance (up to 3 reinfection events, n = 7), including some with divergent genotypes (Table 1).

To determine if the breadth of HCV neutralisation in plasma was associated with clearance of HCV reinfection, a panel of 11 heterologous HCV pseudoparticles (HCVpp) and six cell culture HCV (HCVcc) systems representing the six major HCV genotypes (1-6) were utilised to derive a 'neutralisation score' as a representation of neutralising breadth in plasma (maximum score of 33 and 18 for HCVpp and HCVcc, respectively, Fig. 1A). Neutralisation scores were determined from a single time point from samples collected prior to reinfection events and within 7 months of the estimated reinfection date. These included the 16 reinfection events that resulted in clearance with sampling at a median of 84 days pre-reinfection episode (IQR 42-159 days), and the eight persistent infection events with sampling at a median of 71 days pre-reinfection (IQR 20-178 days). The second comparison was performed during the reinfection event at a median of 84 days post reinfection (IQR 54-160 days) in 16 cleared reinfection events, and at a median of 104 days post reinfection (IQR 42-197 days) for eight persistent reinfection events. Contrary to our hypothesis, neutralising breadth both prior to, and during, reinfection did not differ between clearance and persistent reinfection outcomes ($p > 0.05$, Fig. 1B). Further, neutralisation scores, both prior to and during reinfection, did not correlate with a decreased duration of viraemia in cleared reinfection events (Fig. 1C). Given that neutralisation of HCV E1/E2 can vary greatly in sensitivity and that this sensitivity has recently been classified into four different tiers,³⁰ an alternative, but overlapping HCVpp panel containing 11 HCVpp representing the four tiers was

Table 1. Patient data summary.

Subject ID	Number of identified infections	Infection outcomes ^a	Infecting subtypes ^b	Estimated duration of infection (days)	Estimated time between infections (days)	Time point prior to reinfection (estimated days prior)	Time point during reinfection (estimated days post)
300062	3	C; C; C	Unk, Unk, 1b	113, 120, 91	444, 935	95, 79	95, 79
300086	4	C; P; C; P	2b, 3a, 3a, 1a	167, 778, 311, >365	815, 589, 378	13, 81, 99	25, 81, 99
300089	2	C; P	1b/3a, 3a	269, >365	685	201	201
300138	3	C; C; P	Unk, 3a, 3a	<180, 356, >365	191, 248	191, 110	191, 110
300144	2	C; C	3a, 1a	352, 364	631	185	185
300152	2	C; C	Unk, 1a	122, 167	1,309	152	152
300155	3	P; C; P	3a, 3a, 3a	762, 261, >365	1,383, 371	105, 215	105, 215
300168	3	C; C; C	1b, 1a, 3a	142, 141, 60	120, 825	78, 25	93, 25
300212	4	C; C; C; C	1b, 1a, 3a, 3a	342, 166, 333, 198	358, 1,022, 489	162, 73, 47	162, 86, 47
300225	2	C; C	Unk, 1b	103, 351	416	87	87
300272	2	C; P	3a, 1a	353, >365	2,164	43	43
300354	3	C; C; C	1a, 1a, Unk	135, 257, 51	380, 882	202, 41	193, 41
400087	2	C; C	1b, 3a	123, 303	49	40	40
0384MX	2	C; P	3a, 3a	136, >365	105	42	189
0984MX	2	C; P	Unk, 3a	136, >365	592	11	42

^aC, clearance (<365 days viraemia); P, persistence (>365 days viraemia).

^bUnk, Unknown subtype identified due to low viral loads.

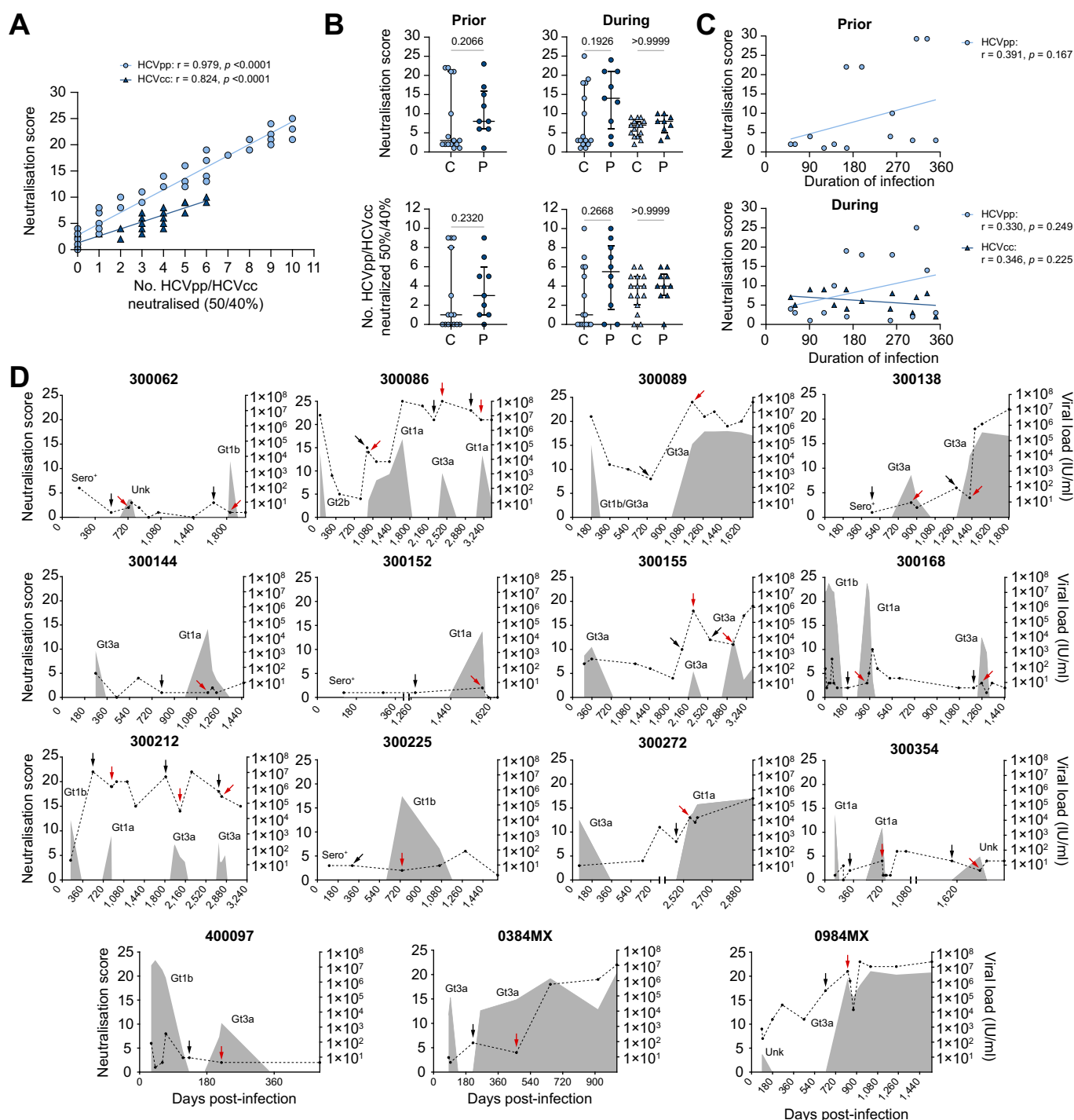


Fig. 1. Analysis of neutralising breadth in HCV reinfection outcome and duration of infection. (A) Linear regression of neutralisation scores and the traditional means for measuring neutralising breadth (No. HCVpp or HCVcc neutralised at 50% and 40% respectively). (B) Comparison of neutralising breadth between cleared reinfections (light blue) and persistent reinfections (dark blue) for both HCVpp (circles) and HCVcc (triangles) immediately prior to reinfection (left) and during reinfection (right) using both the neutralising scores (top) and traditional means (bottom) was performed with a Mann-Whitney U test. (C) Linear regression of neutralisation scores with the estimated duration of infection in cleared reinfections. (D) Longitudinal analysis (days post infection, X axis) of neutralisation scores (black dotted line, left Y axis) and viral load in plasma (grey shading, right Y axis) in all 15 individuals. The infecting Gt is indicated. Sero+ indicates time point at which individuals tested positive for the commercial diagnostic anti-HCV antibody assay but did not have detectable HCV RNA. Unk (unknown) indicates that the genotype for this infection could not be determined due to too low viral load. The black arrows indicate the time point used for comparing 'prior to' reinfection and the red arrows indicate the time point used for comparing 'during' reinfection. Gt, genotype; HCVcc, cell culture HCV; HCVpp, HCV pseudoparticle. (This figure appears in color on the web.)

used to further test if there was an association between neutralisation breadth and reinfection outcome (Fig. S1). At pre-reinfection, no significant differences were detected in any of the tiers. During reinfection, persistent infections were found to neutralise significantly more HCVpp in tier 4 than cleared reinfections ($p = 0.039$, Fig. S1A), specifically, UKN4.11.1 (tier 4) and UKN5.14.4 (tier 4) ($p < 0.05$, Fig. S1B).

Neutralisation scores were also examined longitudinally by individual. Repeated clearance of HCV infection (defined as clearance of two or more infections, $n = 9$) was associated with low neutralisation scores in all but one individual (300212) (Fig. 1D). This individual developed potent neutralisation immediately after clearance of primary infection, which was then sustained longitudinally and was associated with clearance of at least three subsequent reinfections from heterogeneous viruses (Fig. 1D). In contrast, two individuals who developed a persistent reinfection (300086 and 300089) were found to have potent neutralising breadth in plasma during the primary infection, which then waned following viral clearance (Fig. 1D). Despite a broad anamnestic neutralising response upon reinfection in both individuals, the reinfection became persistent (Fig. 1D). For the other individuals who developed a persistent secondary infection, neutralisation scores increased early in reinfection (approximately 110 days post reinfection), and yet viraemia was found to persist. Taken together, this data did not support an association between neutralising breadth and clearance of HCV reinfection.

Outcome of HCV reinfection is not determined by virus-specific neutralisation, neutralising epitope availability or epitope specificities

As neutralising breadth was generally not found to be a determinant of reinfection outcome, where possible, autologous E1/E2 proteins from reinfection were examined to see if reinfection outcomes could be determined by virus-specific neutralisation or epitope availability. Of the 24 reinfection events, 6 E1/E2s from cleared reinfections and 8 E1/E2s from persistent reinfections were able to be sequenced and a dominant sequence was found (Table 2). After cloning and expression in mammalian cells, 8/14 re-infecting E1/E2s were found to bind to a panel of nine well-characterised, representative monoclonal antibodies (mAbs) targeting the known antigenic regions (AR1-5) and domains (domains A-E) within E1/E2 (Table S1), suggesting that these proteins were folding properly. From these eight re-infecting E1/E2s, six were found to produce infectious HCVpp (2 cleared reinfections and 4 persistent reinfections, Table 2). To explore whether pre-existing virus-specific neutralisation, or early-developed virus-specific neutralisation, could determine reinfection outcome, neutralising titres (inhibitory dilution of plasma required to neutralise 50% of the HCVpp expressing autologous E1/E2 [ID_{50}]) were determined for each respective individual, both prior to and during reinfection. Although the sample size was limited, no association between autologous neutralising titres and reinfection outcome was found (Fig. 2A). To expand this analysis to include more autologous E1/E2s, the binding potentials (BP, maximal binding [B_{max}]/dissociation constant [K_d]) of all viable E1/E2s was determined using a panel of six representative neutralising mAbs (Table S1) and the average of these BP from all the mAbs was compared between reinfection

Table 2. Testing autologous E1/E2 from HCV reinfections.

Subject ID	Number of identified infections	Autologous E1/E2 consensus sequence obtained (infection numbers)	Viable after expression in HEK293T cells	Infectious in HCVpp	Average binding potentials (B_{max}/K_d)	Time point of autologous sequence (days post infection)
300062	3	None				
300086	4	2 and 4	2 and 4	2 and 4	4.2 and 28.1	167 and 99
300089	2	1 (Gt1b) and 2	None			
300138	3	2 and 3	2 and 3	None	3.3 and 8.9	191 and 110
300144	2	2 only	2 only	None	4.4	185
300152	2	None				
300155	3	3 only	None			
300168	3	1 and 2	1 and 2	1 and 2	15.2 and 38.7	32 and 93
300212	4	1, 2 and 3	1 and 2	1 and 2	9.9 and 11.3	146 and 162
300225	2	None				
300272	2	2 only	2 only	2 only	15.3	43
300354	3	None				
400087	2	1 and 2	None			
0384MX	2	1 and 2	1 and 2	1 and 2	11.6 and 20.6	94 and 42
0984MX	2	2 only	None			

Gt, genotype.

outcomes (Fig. 2B). No significant difference was found between cleared ($n = 3$) and persistent ($n = 6$) E1/E2 reinfection outcomes ($p = 0.7302$), suggesting that the ability for an Ab to bind to these neutralising epitopes is also not a predictor of reinfection outcome. Further, the average BP from reinfection consensus E1/E2s was not significantly correlated with the duration of viraemia in cleared reinfections ($n = 8$) (Fig. 2B).

No association between antibody epitope responses and infection outcome

To determine if the epitope specificities of the Abs in plasma were associated with reinfection outcome, epitope mapping was conducted using a competition-based ELISA involving the panel of 9 mAbs described above. Mapping was performed using recombinant H77 E1/E2 (genotype 1a) (Fig. 2C and Fig. S2). Mapping was only performed on plasma that had a binding optical density (OD_{450}) to the H77 E1/E2 of 0.7 or more. To determine the hierarchy of the epitopes targeted, an epitope mapping score was determined as previously described.³¹ When the epitope mapping scores were compared between reinfection outcomes, it was found that cleared reinfections had significantly higher AR5-like Abs mapped ($p = 0.0462$, Fig. 2D). However, as there was a high level of conservation of the epitope hierarchy when examined longitudinally, it was possible that this finding was only significant due to the representation of multiple cleared infections from individual 300212. When the same comparison was done using secondary infections only, thus representing each individual only once, this association was not significant (Fig. 2C).

Class-switched HCV E2-specific memory B cell frequencies are associated with reinfection outcomes

To explore if B-cell characteristics associated with reinfection outcome, HCV E2-specific B cells from PBMCs were phenotyped ($CD19+CD20+CD10-IgD-IgG+E2+$) and compared between reinfection outcomes (Fig. S3). Of the 15 individuals with documented reinfection, 3 (300354, 0348MX and 300062) were excluded as their plasma antibodies did not bind above background levels to the E2 tetramers. The remaining 12 individuals represented a mix of reinfection outcomes ($C = 11$, $P = 6$). Only a single time point was used to represent prior and at reinfection for each infection (Fig. S4). E2-specific B cells were divided into naïve (IgD^+) and class-switched (IgD^-) phenotypes and measured longitudinally (Fig. 3A and Fig. S4). Frequencies were greater than the previously reported healthy control background level (0.049%) for all time points prior to reinfection (Fig. S5B).³² IgG^+ E2-specific B cells were further phenotyped into activated memory ($CD21^+$, $CD27^+$), resting memory (RM, $CD21^+$, $CD27^-$), intermediate memory ($CD21^+$, $CD27^-$) and tissue-like memory ($CD21^-$, $CD27^-$) B cells, and CD95 expression was also examined within these subsets (Fig. S5G–J). An initial principal component analysis on the frequencies of these subsets (32 subsets when analysed based on a combination of the following expression of IgD , IgG , $CD21$, $CD27$ and $CD95$) at the time point prior to reinfection, but not during reinfection, showed that the E2-specific B cells formed two distinct phenotypic clusters based on infection outcome (Fig. 3B and Fig. S5A). Analysis of the different subsets showed that the

persistent reinfection outcome, when analysed prior to reinfection, was associated with a significantly higher number of IgG^+ E2-specific B cells ($p = 0.0091$, Fig. 3C and Fig. S5C). More specifically, a significant difference was observed in the RM B-cell subset prior to reinfection ($p = 0.0032$) (data not shown). No significant difference in these two subsets was observed during the reinfection event (Fig. 3E and Fig. S5D). Longitudinal analyses indicated that cleared reinfections were associated with a significant increase in IgG^+ E2-specific B-cell frequencies following reinfection ($p = 0.027$, Fig. 3E and Fig. S5E), whereas persistent reinfections were not ($p = 0.578$, Fig. 3F and Fig. S5F).

As CD95 has been reported to be a marker for apoptosis and B-cell exhaustion,^{33–35} differences in the frequency of $CD95^+$ E2-specific B cells were compared between reinfection outcomes. Prior to reinfection, the frequency of IgG^+ $CD95^+$ E2-specific B cells was found to be significantly higher prior to persistent reinfections when compared to cleared reinfections ($p = 0.0078$, Fig. S5G). Again, this difference was greater in the RM B-cell population ($p = 0.0045$, Fig. S5I). By contrast, the frequency of IgG^+ $CD95^+$ E2-specific B cells was not significantly different between cleared and persistent outcomes during reinfection (IgG^+ E2-specific B cells $p = 0.131$ and RM B cells $p = 0.170$) (Fig. S5H,J).

Transcriptional profiling of HCV E2-specific B cells reveals differences by outcome prior to HCV reinfection

To further explore potential differences in B-cell characteristics between cleared and persistent reinfection outcomes, single-cell transcriptomics was performed on 8 of the 12 individuals analysed by flow cytometry (Table 3). This included a total of 14 reinfections ($C = 8$, $P = 6$) with the same two time points: prior to reinfection and during reinfection. Overall, 1,863 single E2-specific B cells were sequenced and this data was combined with previously published³⁶ single-cell RNA sequencing (scRNAseq) data for 438 receptor binding domain (RBD)-specific B cells (from 3 mild, 1 moderate and 3 severe SARS-CoV-2 convalescent individuals) and 63 non-antigen-specific B cells from two healthy donors. In total, 1,340 cells were retained after quality control and integration (Table 3). Using an unsupervised approach, six clusters were identified for the 15 individuals (Fig. S6A–C). We identified 407 differentially expressed genes (DEGs), with a fold change >0.25 and adjusted p value <0.05 , across the six clusters (Fig. S6D, Tables S8). The transcriptomic data and phenotype data were analysed in combination for specific marker genes³⁷ and phenotypes resulting in the assignment of the following B-cell subsets: activated (act) MBC1, MBC1, MBC2, actMBC2, double-negative (DN)1 and DN2 (Fig. S6D). The first cluster, actMBC1, showed upregulation of genes associated with activation and differed to actBC2 which had increased expression of genes associated with activation and proliferation. actBC1 was dominant in the prior to reinfection time point in the persistence group (48.9%), whereas actBC2 was dominant in the prior to reinfection time point in the clearance group (55.55%) and RBD-specific MBCs from severe SARS-CoV-2 infection (25.86%) (Fig. S6E). MBC1 was generally low in the HCV E2 MBCs but dominant in the RBD-specific MBCs from mild SARS-CoV-2 (68.25%) infection. The third cluster, MBC2 (C2) had low expression of activation

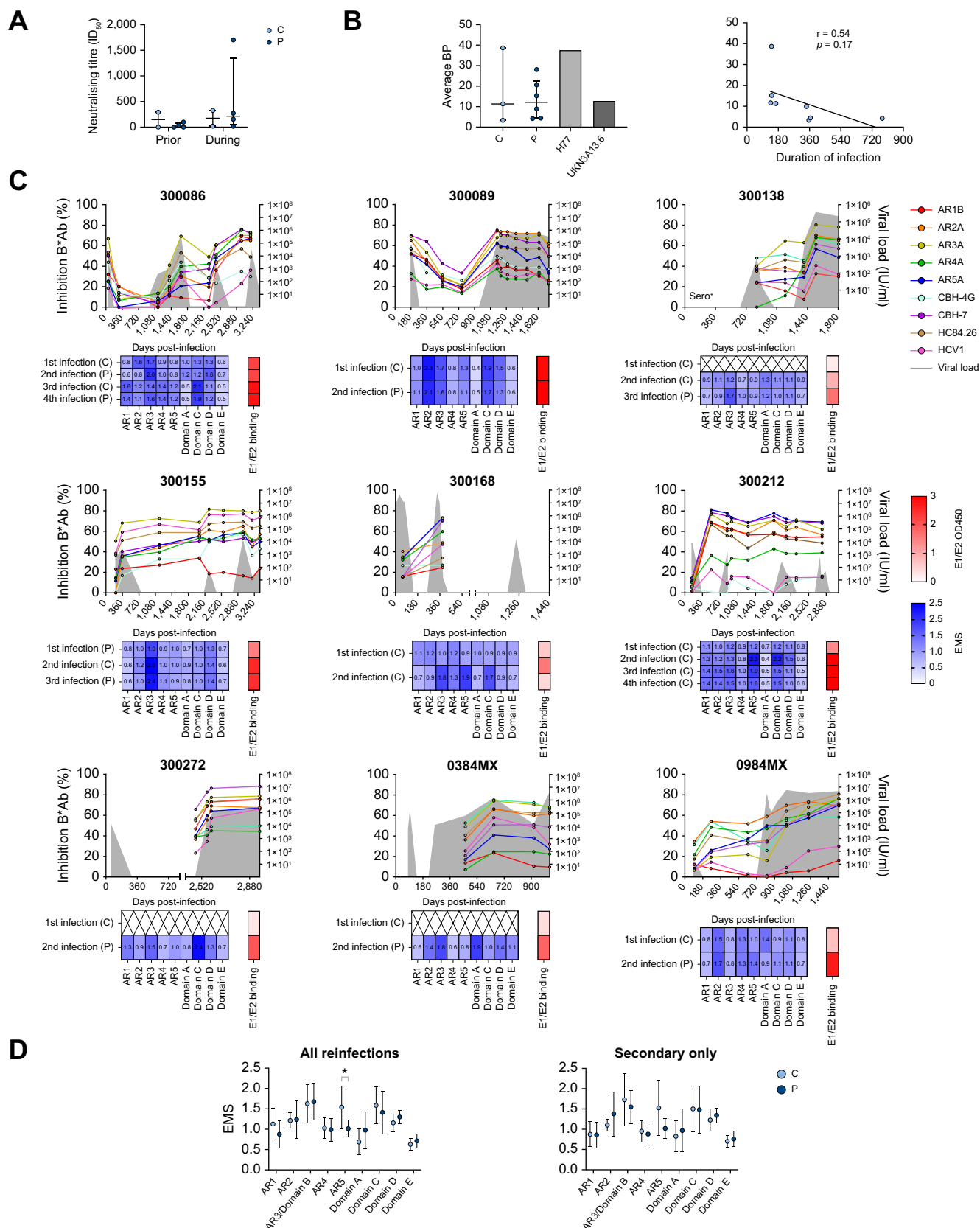


Fig. 2. Analysis of autologous E1/E2s via HCVpp neutralisation and binding potentials (B_{max}/K_D). (A) Comparison of neutralising titres (50% inhibitory dilution [ID₅₀]) against HCVpp expressing autologous E1/E2s from consensus sequences from cleared (light blue, C) and persistent (dark blue, P) reinfections, both prior (left) and during (right) reinfection. (B) Comparison of BPs between cleared (blue, C) and persistent (red, P) reinfection outcomes (Mann-Whitney *U* test, left) with reference BPs to the H77 (Gt1a) and UKN3A13.6 (Gt3a). Linear regression analysis of BPs and duration of infection in those that cleared their reinfection (*i.e.*, regardless of time to

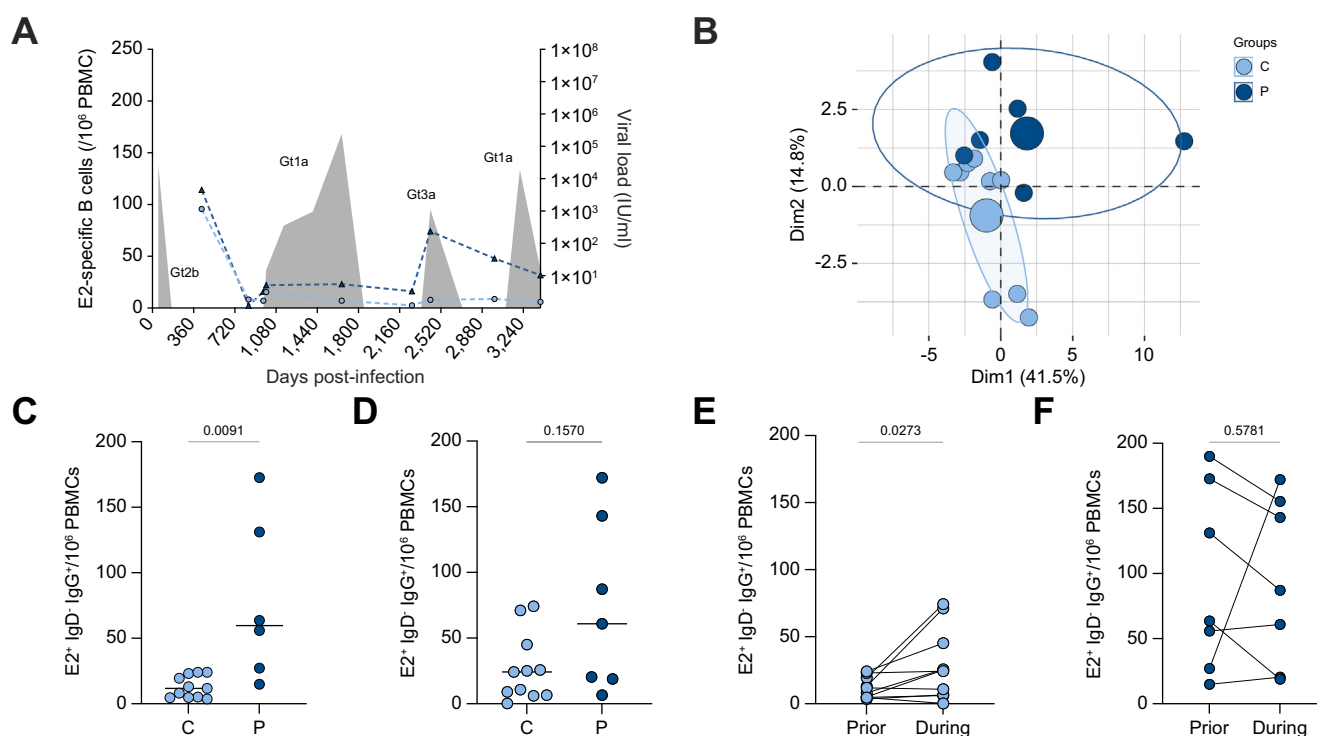


Fig. 3. Longitudinal analysis of E2-specific B cells. (A) Representative longitudinal analysis (days post primary infection, X axis) of naïve (IgD^+ , light blue) and class-switched (IgD^- , dark blue) E2-specific B cells in individual 300086 with multiple infection events. Viral load is represented by the grey shading (right Y axis). The Gt of each infection is shown above the grey shading for each respective infection. (B) Principal component analysis of the E2-specific B-cell subsets analysed at the pre-reinfection time point and separated based on infection outcome, cleared (light blue, C) and persistent (dark blue, P). (C) Comparison, with a linear mixed effects model, of the $\text{E2}^+\text{IgD}^+\text{IgG}^+$ -specific B-cell frequency within the total PBMC population between clearance and persistent at the pre-reinfection time point, and (D) at (during) the reinfection time point. (E) Analysis of the change in $\text{E2}^+\text{IgD}^+\text{IgG}^+$ frequency at the pre- and during reinfection time points in cleared and (F) persistent outcomes. Gt, genotype; PBMC, peripheral blood mononuclear cell.

markers and was dominant in persistent reinfection both at the prior to (32.06%) and during reinfection (31.5%) time points. The last two clusters contained a low number of cells and reported gene signatures similar to the previously described MBCs: Cmem2 and DN2.³⁷ In summary, the HCV E2-specific B cells exhibited different proportions of B-cell subsets to SARS-CoV-2 and this was particularly evident at the prior to reinfection time point between cleared and persistent reinfection outcomes.

To further explore differences between the HCV E2-specific B cells from cleared and persistent reinfections, the HCV E2-specific B cells were analysed independently using unsupervised clustering, resulting in the generation of five clusters (Fig. 4A) across the two outcomes (Fig. 4B). These clusters expressed a similar number of genes, with an average gene expression of 1,625 genes (range: 1,325–2,255), with cluster 4 expressing the highest number of genes. A comparable number of genes were expressed both prior to (median = 1,753) and during reinfection (median = 1,560). DEG analysis identified 331 genes that were differentially expressed across the five distinct

clusters with an adjusted p value <0.05 and average log2 fold change >0.1 .

The five transcriptomic clusters identified for the HCV E2-specific MBCs could be characterised by specific genes and pathways (Fig. S4B, Tables S2 and S3). The first cluster, MBC1, showed upregulation of genes associated with the UV response pathway. Prior to reinfection, this cluster was dominant in persistent reinfections. The second cluster was enriched for TNF- α signalling via the NF- κ B pathway and with genes associated with an activated MBC phenotype previously reported in malaria³⁸ and influenza,³⁹ thus this cluster was termed 'activated memory B cell 1' (actBC1). This actBC1 cluster was dominant in cleared reinfection outcomes prior to reinfection. The third cluster, termed 'inhibitory memory B cell cluster' was enriched with genes in the oxidative phosphorylation pathway and with genes, such as CD52, which have been associated with inhibition of B-cell signalling.⁴⁰ This cluster was present in higher proportions in persistent reinfections at both time points when compared to cleared reinfection outcomes. The fourth and fifth clusters, termed 'memory B cell cluster 2'

clearance) (right, $n = 8$). (C) Longitudinal analysis of epitope specificities in plasma using a competition-based ELISA with nine representative mAbs (see S Table 1) and H77 E1/E2 (Gt1a). % inhibition (left Y axis) of each mAb is represented by a different color (shown on the right) and the viral loads (right Y axis) are shown in grey. Heatmap of each infection shows the epitope hierarchy as an EMS (blue), whereby the higher the score, the higher the targeting of that epitope. Crosses indicate that the binding to the H77 E1/E2 was too low for epitope mapping to be completed. The red heatmap indicates the level of E1/E2 binding (OD450) for each respective time point. (D) Comparison of EMS scores (2-way ANOVA, corrected with Sidak's multiple comparisons test) for each epitope between cleared (light blue, C) and persistent (dark blue, P) reinfection outcomes for all possible reinfections (left) and only secondary infections (right). * $p < 0.05$. BPs, binding potentials; EMS, epitope mapping score; Gt, genotype; HCVpp, HCV pseudoparticle; mAbs, monoclonal antibodies. (This figure appears in color on the web.)

Table 3. Individuals and infections selected for E2-specific B-cell analysis and the total number of identified BCR sequences.

Subject ID	Number of identified infections	Infection outcomes ^a	Time points selected for B-cell analysis (infection numbers)	Total no. E2-specific B cells sorted	Total no. E2-specific B cells retained after QC	No. Identified BCRs	Success rate (%)
300086	4	C, P, C, P	2, 3 and 4	216	95	166	76.9
300089	2	C, P	1 and 2	436	168	367	84.2
300155	3	P, C, P	1, 2 and 3	242	136	176	72.7
300168	3	C, C, C	1, 2 and 3	134	84	125	93.3
300212	4	C, C, C, C	2, 3 and 4	316	134	262	82.9
300272	2	C, P	1 and 2	206	73	197	95.6
400087	2	C, C	1 and 2	72	56	57	79.2
0984MX	2	C, P	1 and 2	241	92	234	97.1

BCR, B-cell receptor.

^aC, clearance (<365 days viraemia); P, persistence (>365 days viraemia).

(MBC2) and 'memory B cell cluster 3' (MBC3), respectively, were represented at lower frequencies in all groups. MBC2 was enriched in fatty acid metabolism and glycolysis pathways. MBC3 was enriched in the estrogen response and coagulation and complement pathways.

To further characterise potential differences between the disease outcome groups, a supervised approach was also performed. Supervised DEG analysis found 93 DEGs across outcome groups prior to reinfection, with an adjusted p value <0.05 (Table S4). Many of the genes highly expressed in the cleared reinfection outcomes were associated with cell activation. Gene set-enrichment analysis (GSEA) of the hallmark pathways identified enrichment of TNF- α signalling via NF- κ B pathway in cleared reinfections at the prior to reinfection time point (Fig. 4C). Genes significantly over-expressed within TNF- α signalling via NF- κ B pathway included: *CD69* ($p = 2.2\text{e-}16$), *DUSP1* ($p = 3.962\text{e-}12$), *FOS* ($p = 1.13\text{e-}13$), *KLF6* ($p = 7.392\text{e-}12$), *CD80* ($p = 0.187$), *CD83* ($p = 2.017\text{e-}10$) and *JUNB* ($p = 3.108\text{e-}12$) (Fig. 4D, Table S5). KEGG pathway analysis showed enrichment of MAPK signalling pathways in cleared reinfections at the prior to reinfection time point, which included the following leading edge genes *FOS*, *DUSP2*, *DUSP1*, *JUN*, *CDC42*, *MAP3K8* and *ATF4*.

In persistent reinfections, DEG analysis showed upregulated expression of *CD52* and human leukocyte antigen genes, such as *HLA-DQA1* and *HLA-DQB1* prior to reinfection (Fig. 4D, Tables S4 and S5). GSEA with the hallmark pathways reported an upregulation of the oxidative phosphorylation pathway and analysis of the KEGG pathways reported enrichment of genes associated with proteasome and antigen processing and presentation pathways, and pathways associated with several disease conditions, including viral myocarditis, systemic lupus erythematosus, type I diabetes mellitus, asthma and autoimmune thyroid disease. Genes involved in these pathways include *HLA-DQB1*, *HLA-DOB*, *H3-3A*, *H4C3*, *HLA-DMB* and *ACTG1* (Table S5).

DEG analysis during reinfection showed only 39 genes that were differentially expressed across outcomes and did not associate with any specific pathways in GSEA at an adjusted p value <0.05. The genes involved in TNF- α signalling via NF- κ B pathways that were significantly different at the prior to reinfection time point showed no significant differences during reinfection (Tables S6 and S7, Fig. 4E).

Soluble CD52 levels correlate with HCV E2-specific B-cell frequency in persistent infection outcome

Surface expressed CD52 and soluble CD52 have been shown to reduce the responsiveness of B cells to B-cell receptor (BCR) engagement.⁴⁰ Given that the observation of increased CD52 expression in HCV E2-specific B cells in persistent reinfections aligned with the observation of limited expansion in B-cell frequency upon reinfection in those with persistent outcomes, soluble CD52 was measured by ELISA for the eight individuals who were analysed by scRNAseq. Although not statistically significant, the level of soluble CD52 prior to reinfection was higher for the persistent reinfection outcomes (Fig. 4F). Also, the level of soluble CD52 in plasma did not show a correlation with mean *CD52* gene expression levels in the E2-specific single B-cell transcriptomic data, though a negative correlation was found between soluble CD52 and E2-specific B-cell

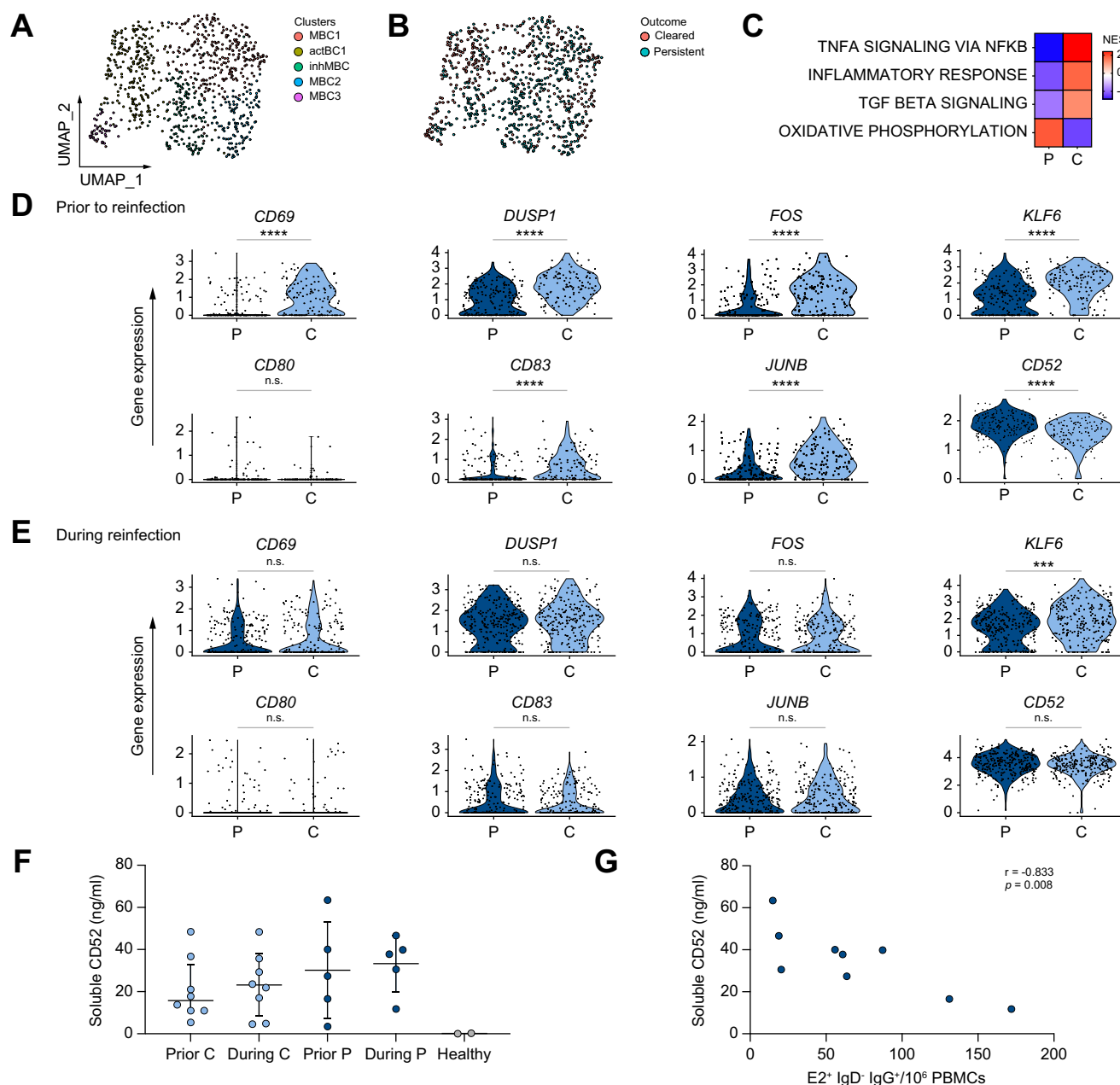


Fig. 4. Transcriptomic analysis of reinfection cohort. (A) UMAP generated using single-cell RNA sequences of E2-specific MBCs across five clusters (MBC1, actBC1, inhMBC, MBC2 and MBC3) and (B) outcome (persistence and clearers). (C) GSEA showing hallmark pathways showing NES across cleared (C) and persistent (P) infection at the pre-reinfection time point. (D) Violin plots of the distribution of genes differentially expressed across cleared (light blue) and persistent (dark blue) pre-reinfection time points and the changes in these genes during (E) reinfection, Mann-Whitney *U* test. (F) Soluble CD52 levels in plasma as measured by ELISA in cleared (light blue) and persistent (dark blue) reinfection outcomes, both prior to and during reinfection. (G) Comparison of the soluble CD52 levels and the frequency of E2⁺IgD⁺IgG⁺ B cells. GSEA, gene set-enrichment analysis; (act/inh)MBCs, (activated/inhibitory) memory B cells; NES, normalized enrichment score; UMAP, uniform manifold approximation projection. (This figure appears in color on the web.)

frequency in persistent reinfections ($r = -0.833$, $p = 0.008$) (Fig. 4G), but not in cleared reinfections.

Antigen-specific B-cell recall and clonal persistence do not associate with HCV reinfection outcome

To determine if there were any differences in the BCR repertoire between the different disease outcomes, the BCRs were analysed longitudinally for these 8 individuals, at time points prior to reinfection and during reinfection. Of the 1,863 HCV E2-

specific B cells sequenced, 1,584 (85.2%) BCRs were successfully reconstructed from the single cells (Table 3). An average of 28% (range 3–55%) of the E2-specific B-cell clones were present in a previous sampling point (referred to here as being recalled), indicating maintenance of previously established MBC responses (Fig. 5A). Further evidence of E2-specific recall was observed when the Ab repertoire, as determined by mass spectrometry, and the B-cell repertoires were compared in individual 300089 (Fig. 5B). At primary infection, 29/90 (32.2%) of the BCR sequences were found as

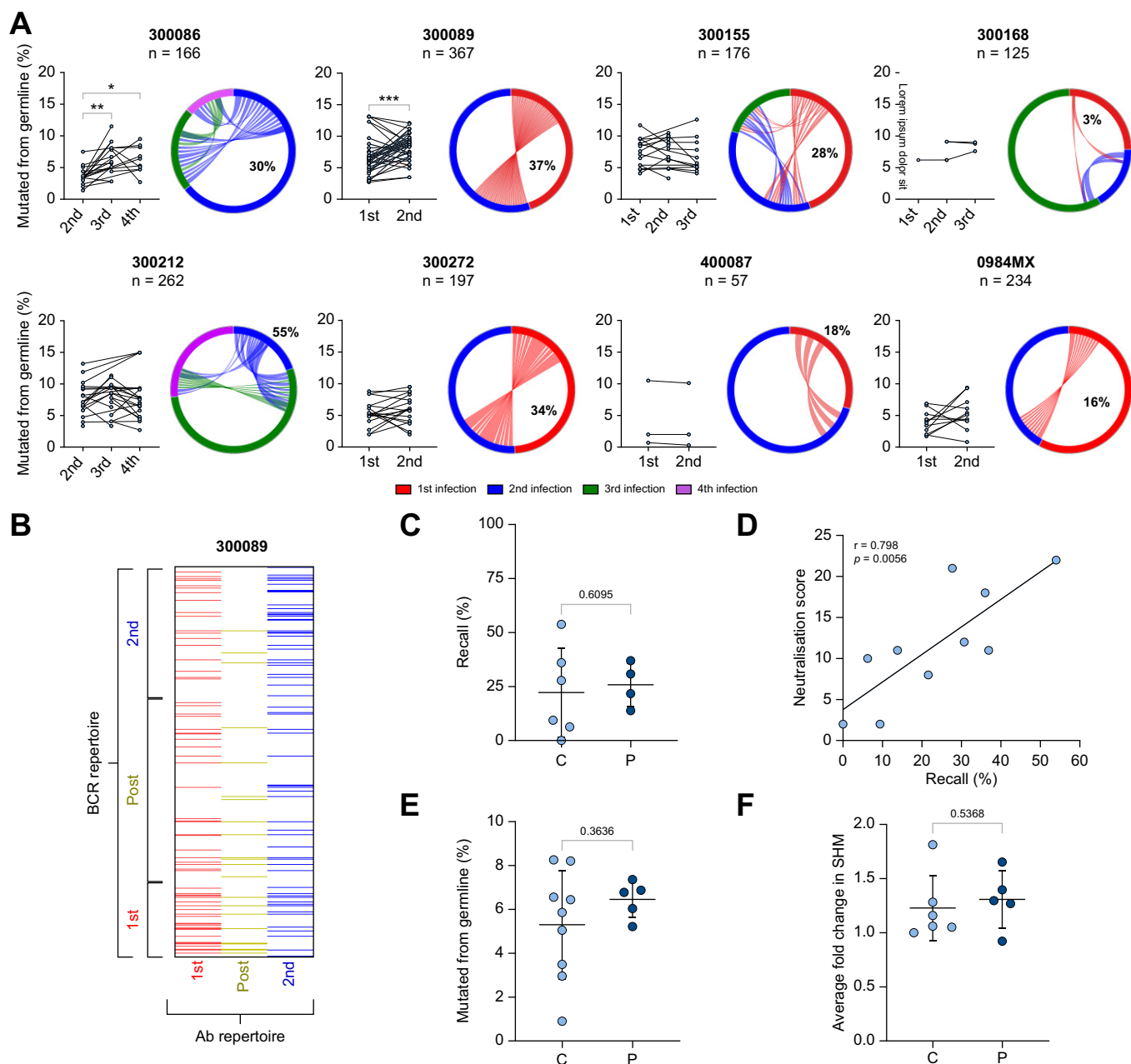


Fig. 5. Longitudinal E2-specific B-cell analysis in individuals with multiple reinfection outcomes. (A) Above each graph is the subject ID with the total number of E2-specific BCR sequences successfully identified (See Table 3). BCR sequences that have been identified at multiple time points are shown in the circus plots, with each ribbon representing recalled clones. The percentage recall from the primary sampled time point is indicated. Each color represents a different infection time point. On the left of each circus plot is the respective recalled BCR sequences in relation to level of mutation from the germline (left Y axis). Comparisons were done using either the Wilcoxon *t* test (2 time points only) or the Kruskal-Wallis test and Dunn's multiple comparison test (3 or more time points). **p* < 0.05, ***p* < 0.01. (B) Heatmap comparison of the BCR and Ab repertoires for individual 300089 at primary infection (1st, red), post primary infection (post, yellow) and secondary infection (2nd, blue). (C) Comparison of the percentage recall (Mann-Whitney *U* test) of E2-specific B cells after the primary sampling time point between cleared (blue, C) and persistent (red, P) reinfection outcomes. (D) Linear regression analysis of the percentage recall of E2-specific B cells after the primary sampling time point and the neutralisation breadth found prior to reinfection. (E) Comparison of the level of mutation from the germline (Mann-Whitney *U* test) between cleared (light blue, C) and persistent (dark blue, P) reinfection outcomes. (F) Comparison of the average fold change in SHM (Mann-Whitney *U* test) between cleared (light blue, C) and persistent (dark blue, P) reinfection outcomes. BCR, B-cell receptor; SHM, somatic hypermutation. (This figure appears in color on the web.)

Abs within plasma, but following clearance of the primary infection, the proportion of BCR sequences that matched plasma Abs was reduced (37/252 [14.7%]), indicating that, while plasma Abs were lost, E2-specific B cells remained in the circulation. At reinfection, the proportion of BCR sequences that matched plasma Abs increased again (40/175 [22.8%]).

indicating that existing memory E2-specific B cells had likely differentiated into antibody-secreting cells.

When the level of B-cell recall was compared between reinfection outcomes (6 cleared reinfections, an average of 22% BCRs were recalled [range 0-54%]; 5 persistent reinfections, average 25% BCRs recalled [range 14-37%])

(Fig. 5C), no significant differences were found ($p = 0.605$). However, the level of recall from the previous infection correlated significantly with neutralisation breadth scores at the prior to reinfection time point ($r = 0.798$, $p = 0.0056$, Fig. 5D).

To explore B-cell recall further, the level of somatic hypermutation (SHM) within B-cell clonotypes was compared across infections and between reinfection outcomes (Fig. 5E,F). However, upon examination of E2-specific B-cell clonotypes across infections, while some individuals (300086 and 300089) had noticeable increases in SHM within their clonotypes at reinfection (Fig. 5A), neither the average level of SHM within B-cell clonotypes, nor the average change in SHM, were significantly different between reinfection outcomes ($p = 0.3636$, $p = 0.5368$ respectively, Fig. 5E,F).

Evidence of antigenic imprinting in HCV infection

To explore recall further, autologous E1/E2s obtained from the primary infection (Table 2) were tested for neutralisation longitudinally and compared to neutralisation against HCVpps carrying autologous E1/E2 from secondary infection. This analysis was possible in two individuals with cleared reinfections (300168, and 300212) and in one with a persistent reinfection (0384MX) (due to limitations in the making of infectious HCVpps). For all three individuals, the neutralisation titre against the HCVpp carrying the primary infection E1/E2 appeared to be boosted preferentially over a response more specific to the HCVpp carrying the secondary infection E1/E2 (Fig. 6). This observation was more prominent in the individual with a persistent reinfection outcome where, despite an ongoing infection from a different virus, the primary E1/E2 continued to be targeted more than the secondary E1/E2.

To understand recall at the antibody level, the epitope mapping of each individual was analysed longitudinally for fluctuations in the epitope hierarchy (Fig. 2C). With the exception of individual 300086, the hierarchy of the epitopes targeted remained conserved across multiple infection events and across persistent infections (Fig. 2C). This indicated that the epitope specificities established during primary infection were maintained during reinfection or persistent infection regardless of infecting genotype or subtype, providing evidence of antigenic imprinting in HCV infection.

Discussion

Identifying correlates of protection in natural immunity is critical to inform vaccine design for genetically diverse RNA viruses. In

this study, 15 individuals who had naturally resolved a primary HCV infection were followed through multiple reinfection events, with natural disease outcomes and the associated immune responses compared. Surprisingly, neither the neutralising breadth in plasma, nor the epitope specificities, were associated with clearance of HCV reinfection. However, the HCV E2-specific B-cell responses did correlate with outcome. In individuals who developed a reinfection that persisted for more than 1 year, a higher level of pre-existing class-switched resting MBCs was found that, unlike the E2-specific B cells in cleared reinfection outcomes, did not expand upon reinfection, and showed higher frequencies of CD95⁺ B cells and expression of genes associated with BCR inhibition. An additional finding of interest in this study was the observation of boosting of primary autologous virus neutralisation titres, and highly conserved epitope specificities between primary and subsequent reinfections, which indicated that HCV may induce 'antigen imprinting' in a similar fashion to SARS-CoV-2 and influenza.^{41,42} Overall, this study suggested that B-cell functionality may be a key consideration for successful protection against highly diverse RNA viruses.

Despite some studies having shown an association of BnAbs with natural clearance of primary HCV infection,^{9,29} and with protection from viral challenge in animal models,^{4,6–8} neither neutralising breadth nor higher levels of virus-specific neutralisation in plasma were associated with reinfection outcomes, although the small sample size in the autologous analysis should be noted. It was noteworthy that of the three individuals who rapidly and repeatedly cleared >3 infections, only one of them generated BnAbs (300212), the other two had narrowly directed nAb responses. In addition, neutralising breadth in plasma was not found to correlate with a reduction in the duration of viraemia in reinfection events. This suggests that neutralisation breadth alone, at the levels measured in our study, was not a determinant of clearance of HCV reinfection. Interestingly, similar results have been reported in other studies, whereby the breadth of neutralisation did not associate with reinfection outcome⁴³ and it has been suggested that a strong cellular response can facilitate clearance in the absence of BnAbs.^{44,45}

With the very recent establishment of a neutralising tiered system for HCV,³⁰ assessment of the neutralising response to the different tiers of HCVpp was done to test whether the ability to neutralise more resistant strains of HCV may be more important than just the ability to neutralise multiple HCV strains. Despite this effort, persistent reinfections were found to have

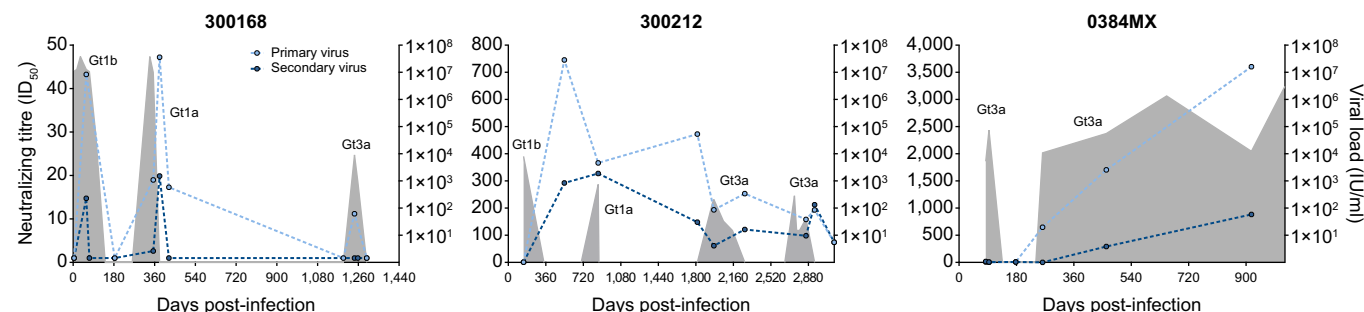


Fig. 6. Neutralisation titres (ID₅₀) against autologous E1/E2s expressed on HCVpps. A longitudinal analysis (days post primary infection, X axis) of neutralising titres (left Y Axis) against HCV expressing autologous E1/E2s from primary infection (light blue) and secondary infection (dark blue) for subjects 300168 (left), 300212 (middle) and 0384MX (right). The viral load is shown in grey (right Y axis) and the infecting genotype (Gt) is shown above the grey shading. HCVpp, HCV pseudoparticle.

higher neutralisation towards the more resistant HCVpp tier (tier 4) during reinfection, which is interesting, and may be something inherent to the properties of the infecting virus. Despite the lack of an association of pre-existing nAb breadth with reinfection outcome, we believe that nAb activity is important for clearance but additional factors are also important, such as the timing of nAb emergence, as we have shown that early emergence of strain-specific nAbs is highly associated with clearance of primary infection.⁴⁶ Unfortunately, limited early sampling of the reinfection event prevented a comparison of the kinetics of nAb emergence with outcome.

The B-cell analysis does support that the humoral responses in the individuals with persistent reinfection were suboptimal. Although the E2-specific B-cell frequencies were lower in the time point prior to cleared reinfections compared to the time point prior to persistent reinfections, the E2-specific B-cell frequencies did significantly expand in the cleared reinfections to comparable levels at reinfection. Phenotypic and transcriptomic profiling indicated that the E2-specific B cells existing prior to persistent reinfection may be dysfunctional, as supported by the observation of a high number of CD95⁺ E2-specific B cells. CD95 is commonly known as an apoptotic or exhaustion marker,^{33–35} although non-antigen-specific B cells expressing CD95 in persistent HCV infection have been reported to be resistant to apoptosis.⁴⁷ Interestingly, HIV ‘elite controllers’ also had significantly lower expression of CD95 on their activated MBCs.³⁵ In this study, scRNAseq identified another marker associated with BCR dysfunction, specifically CD52.⁴⁰ Engagement of membrane bound CD52 has been shown to inhibit BCR signalling,⁴⁰ whereas soluble CD52 in B cells has been shown to inhibit Toll-like receptor and TNF receptor signalling to limit activation of NF- κ B. The observation of increased CD52 RNA expression in persistent infection outcomes was concurrent with the observation of decreased expression of the NF- κ B pathway and inflammatory pathways. Genes associated with the NF- κ B pathway have an important role in cell proliferation and their increased expression in the E2-specific MBCs prior to clearance suggests these cells may be primed and ready to respond. In addition, the E2-specific B cells prior to clearance had increased CD83 expression which has been shown to be important for early induction of anti-influenza antibodies in mice.⁴⁸ In summary, although it is likely that multiple mechanisms of dysfunction contribute to failed protection, this study suggests that E2-specific MBCs’ capacity to respond to the new infection is likely an important factor for subsequent clearance events.

Overall a high level of clonality in the E2-specific MBC repertoire was observed across the time points, with an increase in SHM in many of the longitudinal clones suggesting

that many MBCs are recruited back to the germinal centres and undergo further selection. This provides a molecular basis behind the observation that multiple reinfection events lead to increased nAb breadth in plasma.^{43,49} However, clones with a longitudinal decrease in SHM were also observed, suggesting that naïve B cells were also being recruited for the secondary immune response which may be important for the generation of strain-specific responses. The strong relationship between the level of BCR recall and neutralising breadth in this study is perhaps not surprising but is interesting as this, to the best of our knowledge, has not been shown before for any virus. Importantly, this finding potentially links neutralising breadth to the level of imprinting that might be expected. Imprinting was further supported by the conservation of the epitope hierarchy over multiple infections, and by the fact that the magnitude of nAb titres towards primary autologous virus consistently exceeded that against the reinfecting virus, suggesting that upon reinfection, there was an anamnestic reinforcement of immune responses towards the primary virus – a concept known as ‘back boosting’.⁵⁰ The limited sample size in the nAb imprinting analysis was due to difficulty in generating infectious HCVpp for the different infections. Future studies in other cohorts would be of great interest to confirm the findings observed in this study.

In the present study, immune responses were analysed within a very rare population, for which there is very limited data available in the current literature. However, the relatively small sample size means further confirmation of these findings is important. Also, as antigen-specific single-cell research is costly and labour intensive, only thousands (rather than hundreds of thousands) of HCV E2-specific B cells and BCRs were sampled. Also, only responses to the core E2 region were analysed and it could be that different B-cell responses target the E1/E2 heterodimer. In addition, successful cloning and expression of autologous E1/E2 proteins from reinfecting viruses was limited due to the lack of robust systems for characterisation of autologous HCV envelope proteins. Despite these limitations, the data presented here shows evidence that HCV is capable of inducing antigen imprinting in a manner consistent with other RNA viruses. While antigen imprinting did not appear to associate with reinfection outcomes, immunogen design against HCV will need to be approached with caution, as the immune response against the primary exposure may influence subsequent immune responses. More importantly, this study highlighted B-cell signatures associated with poor reinfection outcome, which may be translatable to other viral infections. Understanding these B-cell signatures will be important for effective vaccine development.

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Abbreviations

ActMBC, activated MBC; BCR, B-cell receptor; BP, binding potentials; DEGs, differentially expressed genes; DN, double-negative; EMS, epitope mapping score; GSEA, gene set-enrichment analysis; HCVcc, cell culture HCV; HCVpp, HCV pseudoparticle; IDU, injecting drug user; MBC, memory B cell; PBMCs, peripheral blood mononuclear cells; SHM, somatic hypermutation.

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Conflicts of interest

The authors declare no conflicts of interest.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

Conceptualisation, R.A.B., A.A.E. & A.R.L. Methodology, A.P.U., B.R.W., A.A.E., I.B., J.J.W., D.A., A.A., N.B. & M.R.W. Software, A.P.U., B.R.W., M.G., A.R. & J.J.W. Validation, A.P.U., B.R.W., I.B., J.J.W., D.A., A.A., N.B. & M.R.W. Formal Analysis, A.P.U., B.R.W., M.G., A.R. & J.J.W. Resources, R.A.B., A.R.L., A.A.E., H.E.D., M.P.D., T.G., J.B. & L.M. Data Curation, A.P.U., M.G. & A.R. Writing - Original Draft, A.P.U. Writing - Review & Editing, A.P.U., R.A.B., A.R.L., H.E.D., M.P.D., J.J.W., J.B. & T.G. Visualisation, A.P.U. B.R.W. & R.A.B. Supervision, R.A.B., A.A.E., A.R.L., F.L., H.E.D., M.P.D. & T.G. Project Administration, R.A.B., A.R.L. & F.L. Funding Acquisition, R.A.B. & A.R.L.

Data availability statement

Single-cell RNA sequencing data have been deposited in GEO with accession number GSE269792. Other metadata, codes and patient specific information is available upon request.

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Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.04.004>.

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